

Test tool qualification for ISO 26262 compliant automotive applications

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Background

Today's complex software based safety critical systems depend on software tools for automation and efficiency. "Tool qualification" is a generic term to describe a process designed to ensure that the risk of a tool error impacting the safety of a system is acceptably low – either because the errors are few, or because they cannot impact safety. Most functional safety standards define processes to achieve tool qualification by considering the application of the tool, and the environment in which it is deployed.

The application considerations are designed to ensure that the tool is used in such a way that potential errors are either avoided or detected. The environmental perspective looks to ensure that the installed tool works as part of the broader tool chain to which it contributes, building confidence and trust in the tool and its use.

All standards have a classification phase to establish the required confidence in the tool and a qualification phase that provides this level of confidence for some or all of its functionality. However the classification and qualification methods differ in some details.

Unlike other IEC 61508 [1] derived standards that also demand a similar level of qualification (Figure 1), the ISO 26262 "Road vehicles - Functional safety" standard [2] prescribes a detailed process for that purpose. This paper discusses that process as it relates to the automotive industry, and references the LDRA tool suite [3] as an example. The LDRA tool suite is an integrated collection of requirements traceability tools, and static and dynamic analysis tools for verification and validation.

Avionics	DO-178C (First published 1992)
Industrial	IEC 61508 (First published 1998, Updated 2010)
Railway	EN 5012X (First published 2001)
Nuclear	IEC 61513 (First published 2001)
Automotive	ISO 26262 (Published 2011)
Medical	IEC 62304 (First published 2006)
Process	IEC 61511 (First published 2003)
...	

Figure 1: Most functional safety standards are direct derivatives of IEC 61508 – that is, part of the "IEC 61058 family"

With the shortfall in advice offered by other standards in mind, the LDRA paper "Test tool qualification for functional safety" discusses in more detail how the tool qualification processes laid out in ISO 26262 might lend themselves to adoption elsewhere.

The ISO 26262 tool qualification process

Despite the derivative nature of the standard, ISO 26262-8 [4] §11 “Confidence in the use of software tools” presents a much more detailed definition of tool qualification than IEC 61508 itself, or most of its other derivatives.

This ISO 26262 supporting process defines a mechanism to provide evidence that the software tool chain can be relied upon. The required level of confidence in a software tool depends upon the circumstances of its deployment, with particular reference in §11.2 to “the possibility that the malfunctioning software tool and its corresponding erroneous output can introduce or fail to detect errors in a safety-related item or element being developed; and the confidence in preventing or detecting such errors in its corresponding output.”

The LDRA tool suite is classified as a TI2 (Tool Impact - 2) category tool because although a verification tool can fail to detect existing errors in source code, unlike (say) an auto code generator it cannot introduce errors into an application.

The TI2 classification dictates that the organization using the tool suite in the development of a compliant application is required to estimate the likelihood of them detecting any error in the tool’s operation (Tool Error Detection), with categories ranging from TD1 (high confidence) to TD3. Evidence of the reasoning behind that assessment is required.

		Tool Error Detection		
		TD1	TD2	TD3
Tool Impact	TI1	TCL1	TCL1	TCL1
	TI2	TCL1	TCL2	TCL3

*Figure 2: Mapping the characteristics of the LDRA tool suite to
“Table 3: Determination of the tool confidence level” from ISO 26262-8:2018*

In turn, the TCL (Tool Confidence Level) is then derived from a look-up table as shown in Figure 2. Depending on the user’s assessment of the application and with reference to its TÜV certification status (below), the resulting TCL will be either TCL1 or TCL2 for the LDRA tool suite. Generally, only TCL2 is realistically achievable.

For all ASILs, tools may be submitted to a validation process in order to confirm their suitability for purpose. However, that is a time consuming and costly approach, and alternatives are offered in the standard for all cases except where the tool suite is assigned TCL2 and the product is designated ASIL D (Figure 2).

Methods		ASIL			
		A	B	C	D
1a	Increased confidence from use in accordance with 11.4.7	++	++	++	+
1b	Evaluation of the tool development process in accordance with 11.4.8	++	++	++	+
1c	Validation of the software tool in accordance with 11.4.9	+	+	+	++
1d	Development in accordance with a safety standard	+	+	+	++
”++” The method is highly recommended for this ASIL. “+” The method is recommended for this ASIL. Applying TCL2 designation					

Figure 3: Assessing the need for the qualification the LDRA tool suite when it is designated TCL2 from “Table 5: Qualification of software tools classified TCL2” from ISO 26262-8:2018

Alternative qualification methods for test tools

ISO 26262 permits the use of alternative qualification methods under some circumstances:

- **Proof in use:** Applicable where a software tool has been used previously for the same purpose with comparable use cases, in similar circumstances (highly recommended for ASIL A-C only)
- **Evaluation of the tool development process** such that evidence can be provided that a suitable software development process has been applied (highly recommended for ASIL A-C only).
- **Development in accordance with a safety standard** where the development of the tool itself complies with a safety standard (highly recommended for ASIL D).

TÜV certification

The second of these alternatives, the evaluation of the tool development process, provides a path followed by many to dramatically reduce the overhead involved with the tool qualification process.



Figure 4: TÜV certification for LDRA tools

The most efficient way to comply with functional safety standards for complex projects is to use process automation, including automated test tools. However, regulatory authorities require that software tools used to automate, reduce or eliminate processes that contribute to the safety lifecycle must be qualified for their intended use. That implies detailed and thorough testing of those tools which is a time consuming and costly process.

In most cases, the most cost-effective approach is therefore to use a tool that is already approved for the applied standard by a TÜV certifying organization.

There are several TÜV certifying organizations. Although they each perform similar services, nevertheless there are sometimes subtle differences in approach between them. LDRA have chosen to secure approval from two completely independent TÜV certifying organizations for their products, leaving no room for doubt from any perspective that best practices have been applied in their development.

ISO 26262 qualification using Tool Qualification Support

A tool validation process is highly recommended by the standard for only the most demanding ASIL D applications where a test tool is assigned TCL 2. However, the fact that the most critical automotive applications demand tool qualification in the context of the nominated development environment suggests that there is potential benefit to adopting a similar approach even for less critical applications – particularly if there is potential for the resulting code to be reused in more critical applications in the future.

In recognition of the need for tool validation, many vendors provide Tool Qualification Support Packs (TQSPs, sometimes known as Tool Qualification Kits) which provide a defined process and associated test cases. Applied correctly, these packs can be used to show whether or not the tool has been configured appropriately to provide the correct results in the tool chain and environment in which it will be deployed.

Referencing the LDRA tool suite as an example, the ISO 26262 TQSP covers three discrete functions, each of which can be specified as an “operational requirement” for the pertinent development project:

- Programming Rules Checking
- Structural Coverage Analysis
- Unit Test / Low Level Test

The TQSP includes five key documents designed to guide the user through the validation process. The process defined by these documents ensure the creation of evidential artefacts and the compilation of reports designed to summarize findings in a form appropriate to the standard.

These five documents are as follows:

- LDRA provide a generic **Tool Criteria Evaluation Report (TCER)** for customization by the user in accordance with the instructions in the TVP. The TCER describes the tool and its architecture, details the certification credit sought, identifies the tool qualification activities to be performed, and summarizes the tool qualification data to be produced.
- The LDRA **Tool Operational Requirements (TOR)** identify one or more of the functions described named above as they relate to the scope of qualification specified by the user in the TCER and SVR.
- The **Tool Verification Plan (TVP)** provided by LDRA is a generic document, common to all projects. It is an instructional script for the verification of the LDRA tool suite. The TVP refers to the TVCP to identify the test cases associated with each of the functions specified in the TOR as requiring qualification for the pertinent project.

- The **Tool Verification Cases and Procedures (TVCP)** provided by LDRA include source code, test cases, and expected results, for use in verifying the effectiveness of the tool on the pertinent installation environment. The user is responsible for ensuring that the test cases contained in the TVCP are sufficient to cover all of the source code constructs to be used on the project.
- **Tool Verification Results (TVR)** are generated by applying the TVCP to test source code in the installation environment. These results are compared with the expected results detailed in the TVCP to establish whether the tool is operating properly under the project's customized operating conditions.

Details of the ISO 26262 tool qualification objectives fulfilled by following this process are detailed in Figure 5.

Part	Section	Description	Document / Life Cycle Data Reference
ISO 26262 Part 8	11.4.3	Compliance Check	TCER
	11.4.4	Planning of usage of a software tool	SVP, TCER, TVP
	11.4.4.1 (a), (b) and (d)	Tool Identification, Environment and Configuration	TCER, TVP, TQR
	11.4.4.1 (c)	Use Cases	TCER, TOR
	11.4.4.2 (a)	Features and Function of Tool	TCER, TVP
	11.4.4.2 (b)	User Manual	TVP
	11.4.4.2 (d), (e) and (f)	Anomalous condition, malfunction (including detection and prevention)	TCER, TOR
	11.4.5	Tool Evaluation	TCER
	11.4.6	Tool Qualification	TCER
	11.4.2	Validity Check of Tool confidence level and Qualification	TCER
	11.4.9	Validation of the software tool	TCER, TVP, TVCP, TVR
	11.4.9.2	Test Cases and Procedures	TVCP
	11.4.9.2	Test Verification Result	TVR
	11.4.10	Confirmation Review	TCER, TQR
	11.5.1	Software tool criteria evaluation report	TCER
	11.5.2	Software tool qualification report	TQR
ISO 26262 Part 6	9.4.5 Table 12 and 9.4.6	Structural coverage metrics at the software unit level and Test Environment	TCER, TVP, TVCP, TVR, TQR
ISO 26262 Part 2	6.4.7 Table 1	Confirmation review of the software tool criteria evaluation report and the software tool qualification report	Review Record
	6.4.7 Table 1	Independence regarding the developers of the item and project management Item 1: the confirmation measure shall be performed by a different person	Review Record

Figure 5: Tool qualification objectives fulfilled through the applications of the LDRA Tool Qualification Support Pack

Conclusions

“Tool qualification” is a generic term to describe a process designed to ensure that the risk of a tool error impacting the safety of a system is acceptably low – either because the errors are few, or because they cannot impact safety. Most functional safety standards including ISO 26262 define processes to achieve tool qualification by considering the application of the tool, and the environment in which it is deployed.

ISO 26262 states that for most ASILs using tools that evidence of acceptable assessment (usually TÜV certification) will be sufficient reassurance that they are fit for purpose. But the fact that the most critical automotive applications demand tool qualification in the context of the nominated development environment suggests that there is potential benefit to adopting a similar approach even for less critical applications.

Many vendors including LDRA provide a collection of documentation including test cases with expected results and reporting processes. Usually known as Tool Qualification Kits or (as in LDRA’s case) Tool Qualification Support Packs, these artefacts can be used to show whether or not a tool has been configured appropriately to provide the correct results in the tool chain in which it will be deployed.

Works cited

- [1] International Electrotechnical Commission (IEC), IEC 61508 Functional safety of electrical/electronic/programmable electronic safety-related systems, IEC, 2010 (Part 1).
- [2] International Organization for Standardization, ISO 26262:2018 “Road vehicles - Functional safety”, International Organization for Standardization, 2018.
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- [4] International Organization for Standardization, in ISO 26262 8:2018 “Road vehicles — Functional safety — Supporting processes”, International Organization for Standardization, 2018.



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